Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY and KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Second Supplemental Testimony

of

Elizabeth D. Arangio

Case 19-G-0309
Case 19-G-0310

Dated: December 13, 2019

Q. Please state your name and business address.

A. My name is Elizabeth D. Arangio. My business address is 40 Sylvan Road, Waltham, Massachusetts 02451.

Q. Are you the same Elizabeth D. Arangio who previously submitted testimony on behalf of KEDNY and KEDLI in these proceedings?

A. Yes, I am. The terms defined in my previous testimonies have the same meanings here.

Q. What is the purpose of your second supplemental testimony?

A. The purpose of my second supplemental testimony is to provide updates to certain exhibits previously submitted in my direct, supplemental, and corrections and updates testimonies. Specifically, the following exhibits were prepared under my direction and supervision:

Exhibit __ (EDA-1SS)	KEDNY/KEDLI Projected Monthly Gas Stored Volumes and Dollars for the Rate Year and Data Years Summarized by Market Area, Gulf Coast and LNG storage;
Exhibit __ (EDA-2SS)	KEDNY/KEDLI Forecast of Variable Gas Expense for the TME March 31, 2021, 2022, 2023 and 2024;
Exhibit __ (EDA-3SS)	KEDNY/KEDLI Forecast of Purchased Gas Expense for the TME March 31, 2021, 2022, 2023 and 2024;

Exhibit __ (EDA-4SS)	KEDNY and KEDLI Estimated Marginal Commodity Cost of Gas;
Exhibit __ (EDA-5SS)	KEDNY and KEDLI Estimated Annualized Marginal Capacity Cost of Gas;
Exhibit __ (EDA-6SS)	The Companies' updated peak day supply/demand.

Q. What updates have been made to the exhibits?

A. The exhibits have been revised based on an updated gas sales forecast provided by Company Witness Theodore Poe, Jr. On November 26, 2019, the Commission issued an "Order Adopting and Approving Settlement" in Case 19-G-0678 ("November 26 Order"). The November 26 Order approved a settlement between the Companies and DPS Staff concerning the Companies' moratorium on new gas connections and provided that the Companies will lift the moratorium on new gas service at least until September 1, 2021. Consequently, the Companies updated their gas sales forecast to account for the addition of new customers. Please see the second supplemental testimony of Company Witness Theodore Poe, Jr. for a further discussion of the updated gas sales forecast.

Further, I updated the forecast fixed costs to include the impacts of recent pipeline rate cases, as well as to account for additional peaking supplies

secured since the Companies' corrections and updates filing on July 3, 2019.

Finally, please note that all the exhibits in this second supplemental filing reflect the Companies' strategy to meet forecast demand without NESE, and thus no NESE fixed or variable costs are included. Exhibit __ (EDA-6SS) reflects the latest peak day supply/demand balance and includes planned CNG and city gate peaking supplies that will be used to supply the forecast demand.

Q. How do the Companies plan to supply the incremental demands created by the increased number of customers?

A. The Companies developed a plan for the next two winters that includes expanded use of portable CNG and increased reliance on city gate peaking supplies. The Companies are also investigating further increases in CNG as needed. The Companies' construction and operation plans associated with the expanded use of CNG are discussed in more detail in the second supplemental testimony of the GIOP. The Companies will continue to procure city gate and CNG supplies to serve peak day and peak hour requirements for as long as they are needed. The planned increase in CNG

volumes, and continued acquisition of city gate peaking supplies, are reflected in Exhibit (EDA-6SS).

Q. Does this conclude your corrections and updates testimony?

A. Yes, it does.

Second Supplemental Direct Testimony of Elizabeth D. Arangio

INDEX OF EXHIBITS

Exhibit __ (EDA-1SS): KEDNY/KEDLI Projected Monthly Gas Stored Volumes and Dollars for the Rate Year and Data Years Summarized by Market Area, Gulf Coast and LNG storage

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Exhibit __ (EDA-3SS): KEDNY/KEDLI Forecast of Purchased Gas Expense for the TME March 31, 2021, 2022, 2023 and 2024

Exhibit __ (EDA-4SS): KEDNY and KEDLI Estimated Marginal Commodity Cost of Gas

Exhibit __ (EDA-5SS): KEDNY and KEDLI Estimated Annualized Marginal Capacity Cost of Gas

Exhibit __ (EDA-6SS): The Companies' updated peak day supply/demand

Second Supplemental Direct Testimony of Elizabeth D. Arangio

Exhibit ___ (EDA-1SS)

KEDNY/KEDLI Projected Monthly Gas Stored Volumes and Dollars for the Rate Year and Data Years Summarized by Market Area, Gulf Coast and LNG storage

Exhibit_(EDA-1SS)
Page 1 of 4

KEDNY / KEDLI

Projected Gas Storage Inventory

Twelve Months Ended March 2021

Market Area	Forecast Apr-2020	Forecast May-2020	Forecast Jun-2020	Forecast Jul-2020	Forecast Aug-2020	Forecast Sep-2020	Forecast Oct-2020	Forecast Nov-2020	Forecast Dec-2020	Forecast Jan-2021	Forecast Feb-2021	Forecast Mar-2021
Dth												
Beginning Inventory	838,479	3,198,405	13,443,799	23,477,646	31,958,090	40,805,023	49,872,594	58,372,709	55,554,911	44,104,438	24,727,009	8,870,552
Injections	2,636,063	10,245,394	10,033,847	8,480,444	8,846,933	9,067,571	8,500,115	688,328	5,712	-	-	4,760
Withdrawals	276,136	-	-	-	-	-	-	3,506,126	11,456,185	19,377,429	15,856,457	8,139,783
EndingBalance	3,198,405	13,443,799	23,477,646	31,958,090	40,805,023	49,872,594	58,372,709	55,554,911	44,104,438	24,727,009	8,870,552	735,528
$												
Beginning Inventory	$ 1,893,665	$ 6,787,185	$ 27,140,729	$ 47,310,789	$ 64,537,746	$ 82,506,460	$ 99,352,543	$ 115,211,665	$ 109,696,203	$ 87,120,235	$ 48,776,286	$ 17,494,039
Injections	$ 5,465,648	$ 20,353,544	$ 20,170,060	$ 17,226,957	$ 17,968,714	$ 16,846,084	$ 15,859,121	$ 1,413,120	$ 13,315	$ -	$ -	$ 11,130
Withdrawals	$ 572,128	$ -	$ -	$ -	$ -	$ -	$ -	$ 6,928,582	$ 22,589,283	$ 38,343,948	$ 31,282,248	$ 16,052,610
EndingBalance	$ 6,787,185	$ 27,140,729	$ 47,310,789	$ 64,537,746	$ 82,506,460	$ 99,352,543	$ 115,211,665	$ 109,696,203	$ 87,120,235	$ 48,776,286	$ 17,494,039	$ 1,452,559
Average Rate	$ 2.1221	$ 2.0188	$ 2.0151	$ 2.0194	$ 2.0220	$ 1.9921	$ 1.9737	$ 1.9746	$ 1.9753	$ 1.9726	$ 1.9721	$ 1.9749

Gulf Coast	Forecast Apr-2020	Forecast May-2020	Forecast Jun-2020	Forecast Jul-2020	Forecast Aug-2020	Forecast Sep-2020	Forecast Oct-2020	Forecast Nov-2020	Forecast Dec-2020	Forecast Jan-2021	Forecast Feb-2021	Forecast Mar-2021
Dth												
Beginning Inventory	2,552,927	3,078,644	3,708,450	3,910,396	3,910,396	3,910,396	4,052,610	4,052,610	4,052,610	4,052,610	4,052,610	4,052,610
Injections	525,718	629,806	201,946	-	-	142,214	-	-	-	-	-	-
Withdrawals	-	-	-	-	-	-	-	-	-	-	-	1,499,683
EndingBalance	3,078,644	3,708,450	3,910,396	3,910,396	3,910,396	4,052,610	4,052,610	4,052,610	4,052,610	4,052,610	4,052,610	2,552,927
$												
Beginning Inventory	$ 6,010,861	$ 7,218,610	$ 8,662,913	$ 9,132,007	$ 9,132,007	$ 9,132,007	$ 9,462,352	$ 9,462,352	$ 9,462,352	$ 9,462,352	$ 9,462,352	$ 9,462,352
Injections	$ 1,207,749	$ 1,444,302	$ 469,094	$ -	$ -	$ 330,345	$ -	$ -	$ -	$ -	$ -	$ -
Withdrawals	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ 3,501,578
EndingBalance	$ 7,218,610	$ 8,662,913	$ 9,132,007	$ 9,132,007	$ 9,132,007	$ 9,462,352	$ 9,462,352	$ 9,462,352	$ 9,462,352	$ 9,462,352	$ 9,462,352	$ 5,960,773
Average Rate	$ 2.3447	$ 2.3360	$ 2.3353	$ 2.3353	$ 2.3353	$ 2.3349	$ 2.3349	$ 2.3349	$ 2.3349	$ 2.3349	$ 2.3349	$ 2.3349

LNG	Forecast Apr-2020	Forecast May-2020	Forecast Jun-2020	Forecast Jul-2020	Forecast Aug-2020	Forecast Sep-2020	Forecast Oct-2020	Forecast Nov-2020	Forecast Dec-2020	Forecast Jan-2021	Forecast Feb-2021	Forecast Mar-2021
Dth												
Beginning Inventory	1,365,902	1,288,022	1,580,146	1,747,191	1,666,715	1,743,881	2,071,001	2,281,673	2,298,000	2,270,880	2,243,760	2,219,265
Injections	-	372,600	244,925	-	157,642	405,000	291,148	42,572	-	-	-	-
Withdrawals	77,880	80,476	77,880	80,476	80,476	77,880	80,476	26,245	27,120	27,120	24,495	27,120
EndingBalance	1,288,022	1,580,146	1,747,191	1,666,715	1,743,881	2,071,001	2,281,673	2,298,000	2,270,880	2,243,760	2,219,265	2,192,145
$												
Beginning Inventory	$ 2,722,867	$ 2,567,732	$ 3,100,110	$ 3,410,882	$ 3,253,576	$ 3,396,339	$ 3,951,874	$ 4,307,170	$ 4,338,857	$ 4,286,976	$ 4,235,095	$ 4,188,235
Injections	$ -	$ 691,597	$ 463,425	$ -	$ 299,812	$ 705,962	$ 508,677	$ 81,891	$ -	$ -	$ -	$ -
Withdrawals	$ 155,134	$ 159,219	$ 152,653	$ 157,307	$ 157,049	$ 150,427	$ 153,381	$ 50,204	$ 51,881	$ 51,881	$ 46,860	$ 51,881
EndingBalance	$ 2,567,732	$ 3,100,110	$ 3,410,882	$ 3,253,576	$ 3,396,339	$ 3,951,874	$ 4,307,170	$ 4,338,857	$ 4,286,976	$ 4,235,095	$ 4,188,235	$ 4,136,355
Average Rate	$ 1.9935	$ 1.9619	$ 1.9522	$ 1.9521	$ 1.9476	$ 1.9082	$ 1.8877	$ 1.8881	$ 1.8878	$ 1.8875	$ 1.8872	$ 1.8869

Exhibit_(EDA-1SS)
Page 2 of 4

KEDNY / KEDLI

Projected Gas Storage Inventory

Twelve Months Ended March 2022

Market Area	Forecast Apr-2021	Forecast May-2021	Forecast Jun-2021	Forecast Jul-2021	Forecast Aug-2021	Forecast Sep-2021	Forecast Oct-2021	Forecast Nov-2021	Forecast Dec-2021	Forecast Jan-2022	Forecast Feb-2022	Forecast Mar-2022
Dth												
Beginning Inventory	735,528	3,142,364	13,296,493	23,330,340	32,745,349	41,426,266	50,391,566	58,551,367	55,554,911	44,394,864	24,914,222	8,828,668
Injections	2,717,024	10,154,129	10,033,847	9,415,008	8,680,917	8,965,301	8,159,800	328,339	2,856	-	-	4,760
Withdrawals	310,188	-	-	-	-	-	-	3,324,795	11,162,903	19,480,643	16,085,553	8,123,976
EndingBalance	3,142,364	13,296,493	23,330,340	32,745,349	41,426,266	50,391,566	58,551,367	55,554,911	44,394,864	24,914,222	8,828,668	709,452
$												
Beginning Inventory	$ 1,452,559	$ 6,386,900	$ 26,341,782	$ 45,728,411	$ 63,939,215	$ 81,002,411	$ 97,763,403	$ 113,234,588	$ 107,463,935	$ 85,905,367	$ 48,118,371	$ 17,026,426
Injections	$ 5,548,247	$ 19,954,882	$ 19,386,630	$ 18,210,803	$ 17,063,196	$ 16,760,991	$ 15,471,185	$ 668,113	$ 6,710	$ -	$ -	$ 11,198
Withdrawals	$ 613,907	$ -	$ -	$ -	$ -	$ -	$ -	$ 6,438,765	$ 21,565,278	$ 37,786,996	$ 31,091,945	$ 15,672,448
EndingBalance	$ 6,386,900	$ 26,341,782	$ 45,728,411	$ 63,939,215	$ 81,002,411	$ 97,763,403	$ 113,234,588	$ 107,463,935	$ 85,905,367	$ 48,118,371	$ 17,026,426	$ 1,365,177
Average Rate	$ 2.0325	$ 1.9811	$ 1.9600	$ 1.9526	$ 1.9553	$ 1.9401	$ 1.9339	$ 1.9344	$ 1.9350	$ 1.9314	$ 1.9285	$ 1.9243

Gulf Coast	Forecast Apr-2021	Forecast May-2021	Forecast Jun-2021	Forecast Jul-2021	Forecast Aug-2021	Forecast Sep-2021	Forecast Oct-2021	Forecast Nov-2021	Forecast Dec-2021	Forecast Jan-2022	Forecast Feb-2022	Forecast Mar-2022
Dth												
Beginning Inventory	2,552,927	3,078,644	3,708,450	4,052,610	4,052,610	4,052,610	4,052,610	4,052,610	4,052,610	4,052,610	4,052,610	4,052,610
Injections	525,718	629,806	344,160	-	-	-	-	-	-	-	-	-
Withdrawals	-	-	-	-	-	-	-	-	-	-	-	1,499,683
EndingBalance	3,078,644	3,708,450	4,052,610	4,052,610	4,052,610	4,052,610	4,052,610	4,052,610	4,052,610	4,052,610	4,052,610	2,552,927
$												
Beginning Inventory	$ 5,960,773	$ 7,183,022	$ 8,639,547	$ 9,447,774	$ 9,447,774	$ 9,447,774	$ 9,447,774	$ 9,447,774	$ 9,447,774	$ 9,447,774	$ 9,447,774	$ 9,447,774
Injections	$ 1,222,248	$ 1,456,525	$ 808,228	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Withdrawals	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ 3,496,184
EndingBalance	$ 7,183,022	$ 8,639,547	$ 9,447,774	$ 9,447,774	$ 9,447,774	$ 9,447,774	$ 9,447,774	$ 9,447,774	$ 9,447,774	$ 9,447,774	$ 9,447,774	$ 5,951,590
Average Rate	$ 2.3332	$ 2.3297	$ 2.3313	$ 2.3313	$ 2.3313	$ 2.3313	$ 2.3313	$ 2.3313	$ 2.3313	$ 2.3313	$ 2.3313	$ 2.3313

LNG	Forecast Apr-2021	Forecast May-2021	Forecast Jun-2021	Forecast Jul-2021	Forecast Aug-2021	Forecast Sep-2021	Forecast Oct-2021	Forecast Nov-2021	Forecast Dec-2021	Forecast Jan-2022	Forecast Feb-2022	Forecast Mar-2022
Dth												
Beginning Inventory	2,192,145	2,114,265	2,246,270	2,250,493	2,170,017	2,089,541	2,277,925	2,290,945	2,298,000	2,261,339	1,803,909	1,610,955
Injections	-	212,482	82,102	-	-	266,264	93,496	33,300	-	-	-	-
Withdrawals	77,880	80,476	77,880	80,476	80,476	77,880	80,476	26,245	36,661	457,431	192,954	36,661
EndingBalance	2,114,265	2,246,270	2,250,493	2,170,017	2,089,541	2,277,925	2,290,945	2,298,000	2,261,339	1,803,909	1,610,955	1,574,294
$												
Beginning Inventory	$ 4,136,355	$ 3,988,510	$ 4,225,531	$ 4,225,972	$ 4,074,159	$ 3,922,345	$ 4,239,729	$ 4,254,790	$ 4,269,219	$ 4,201,094	$ 3,369,929	$ 3,018,535
Injections	$ -	$ 389,512	$ 147,577	$ -	$ -	$ 463,593	$ 165,234	$ 63,605	$ -	$ -	$ -	$ -
Withdrawals	$ 147,845	$ 152,492	$ 147,136	$ 151,813	$ 151,813	$ 146,209	$ 150,172	$ 49,177	$ 68,125	$ 831,165	$ 351,394	$ 68,125
EndingBalance	$ 3,988,510	$ 4,225,531	$ 4,225,972	$ 4,074,159	$ 3,922,345	$ 4,239,729	$ 4,254,790	$ 4,269,219	$ 4,201,094	$ 3,369,929	$ 3,018,535	$ 2,950,410
Average Rate	$ 1.8865	$ 1.8811	$ 1.8778	$ 1.8775	$ 1.8771	$ 1.8612	$ 1.8572	$ 1.8578	$ 1.8578	$ 1.8681	$ 1.8738	$ 1.8741

Exhibit_____(EDA-1SS)
Page 2 of 4

Exhibit_(EDA-1SS)
Page 3 of 4

KEDNY / KEDLI

Projected Gas Storage Inventory

Twelve Months Ended March 2023

Market Area	Forecast Apr-2022	Forecast May-2022	Forecast Jun-2022	Forecast Jul-2022	Forecast Aug-2022	Forecast Sep-2022	Forecast Oct-2022	Forecast Nov-2022	Forecast Dec-2022	Forecast Jan-2023	Forecast Feb-2023	Forecast Mar-2023
Dth												
Beginning Inventory	709,452	3,088,380	13,330,695	23,364,542	32,793,005	41,407,258	50,368,482	58,550,380	55,554,911	44,055,027	24,456,916	8,204,020
Injections	2,699,133	10,242,315	10,033,847	9,428,463	8,614,253	8,961,224	8,181,897	311,032	5,712	-	-	3,808
Withdrawals	320,206	-	-	-	-	-	-	3,306,500	11,505,595	19,598,112	16,252,896	7,370,358
EndingBalance	3,088,380	13,330,695	23,364,542	32,793,005	41,407,258	50,368,482	58,550,380	55,554,911	44,055,027	24,456,916	8,204,020	837,470
$												
Beginning Inventory	$ 1,365,177	$ 6,244,998	$ 25,727,568	$ 45,084,165	$ 63,328,555	$ 80,072,734	$ 96,968,793	$ 112,561,486	$ 106,829,682	$ 84,767,243	$ 46,962,425	$ 15,736,682
Injections	$ 5,494,731	$ 19,482,570	$ 19,356,596	$ 18,244,391	$ 16,744,178	$ 16,896,060	$ 15,592,693	$ 641,338	$ 13,497	$ -	$ -	$ 8,978
Withdrawals	$ 614,910	$ -	$ -	$ -	$ -	$ -	$ -	$ 6,373,144	$ 22,075,934	$ 37,804,818	$ 31,225,743	$ 14,129,848
EndingBalance	$ 6,244,998	$ 25,727,568	$ 45,084,165	$ 63,328,555	$ 80,072,734	$ 96,968,793	$ 112,561,486	$ 106,829,682	$ 84,767,243	$ 46,962,425	$ 15,736,682	$ 1,615,813
Average Rate	$ 2.0221	$ 1.9299	$ 1.9296	$ 1.9312	$ 1.9338	$ 1.9252	$ 1.9225	$ 1.9230	$ 1.9241	$ 1.9202	$ 1.9182	$ 1.9294

Gulf Coast	Forecast Apr-2022	Forecast May-2022	Forecast Jun-2022	Forecast Jul-2022	Forecast Aug-2022	Forecast Sep-2022	Forecast Oct-2022	Forecast Nov-2022	Forecast Dec-2022	Forecast Jan-2023	Forecast Feb-2023	Forecast Mar-2023
Dth												
Beginning Inventory	2,552,927	3,078,644	3,708,450	3,911,613	3,911,613	3,911,613	4,052,610	4,052,610	4,052,610	4,052,610	4,052,610	4,052,610
Injections	525,718	629,806	203,163	-	-	140,997	-	-	-	-	-	-
Withdrawals	-	-	-	-	-	-	-	-	-	-	-	1,499,683
EndingBalance	3,078,644	3,708,450	3,911,613	3,911,613	3,911,613	4,052,610	4,052,610	4,052,610	4,052,610	4,052,610	4,052,610	2,552,927
$												
Beginning Inventory	$ 5,951,590	$ 7,201,762	$ 8,694,313	$ 9,182,836	$ 9,182,836	$ 9,182,836	$ 9,521,875	$ 9,521,875	$ 9,521,875	$ 9,521,875	$ 9,521,875	$ 9,521,875
Injections	$ 1,250,172	$ 1,492,550	$ 488,524	$ -	$ -	$ 339,039	$ -	$ -	$ -	$ -	$ -	$ -
Withdrawals	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ 3,523,605
EndingBalance	$ 7,201,762	$ 8,694,313	$ 9,182,836	$ 9,182,836	$ 9,182,836	$ 9,521,875	$ 9,521,875	$ 9,521,875	$ 9,521,875	$ 9,521,875	$ 9,521,875	$ 5,998,270
Average Rate	$ 2.3393	$ 2.3445	$ 2.3476	$ 2.3476	$ 2.3476	$ 2.3496	$ 2.3496	$ 2.3496	$ 2.3496	$ 2.3496	$ 2.3496	$ 2.3496

LNG	Forecast Apr-2022	Forecast May-2022	Forecast Jun-2022	Forecast Jul-2022	Forecast Aug-2022	Forecast Sep-2022	Forecast Oct-2022	Forecast Nov-2022	Forecast Dec-2022	Forecast Jan-2023	Forecast Feb-2023	Forecast Mar-2023
Dth												
Beginning Inventory	1,574,294	1,508,774	1,556,290	1,644,837	1,577,133	1,509,429	1,659,909	1,673,910	1,680,000	1,652,918	1,625,837	1,601,376
Injections	-	115,219	154,067	-	-	216,000	81,705	32,298	-	-	-	-
Withdrawals	65,520	67,704	65,520	67,704	67,704	65,520	67,704	26,208	27,082	27,082	24,461	27,082
EndingBalance	1,508,774	1,556,290	1,644,837	1,577,133	1,509,429	1,659,909	1,673,910	1,680,000	1,652,918	1,625,837	1,601,376	1,574,294
$												
Beginning Inventory	$ 2,950,410	$ 2,827,618	$ 2,901,841	$ 3,053,282	$ 2,927,604	$ 2,801,927	$ 3,060,063	$ 3,080,243	$ 3,093,824	$ 3,043,952	$ 2,994,080	$ 2,949,033
Injections	$ -	$ 200,841	$ 273,365	$ -	$ -	$ 379,340	$ 144,870	$ 61,829	$ -	$ -	$ -	$ -
Withdrawals	$ 122,792	$ 126,618	$ 121,924	$ 125,678	$ 125,678	$ 121,203	$ 124,691	$ 48,247	$ 49,872	$ 49,872	$ 45,046	$ 49,872
EndingBalance	$ 2,827,618	$ 2,901,841	$ 3,053,282	$ 2,927,604	$ 2,801,927	$ 3,060,063	$ 3,080,243	$ 3,093,824	$ 3,043,952	$ 2,994,080	$ 2,949,033	$ 2,899,161
Average Rate	$ 1.8741	$ 1.8646	$ 1.8563	$ 1.8563	$ 1.8563	$ 1.8435	$ 1.8401	$ 1.8416	$ 1.8416	$ 1.8416	$ 1.8416	$ 1.8416

Exhibit_____(EDA-1SS)
Page 3 of 4

Exhibit_(EDA-1SS)
Page 4 of 4

KEDNY / KEDLI

Projected Gas Storage Inventory

Twelve Months Ended March 2024

Market Area	Forecast Apr-2023	Forecast May-2023	Forecast Jun-2023	Forecast Jul-2023	Forecast Aug-2023	Forecast Sep-2023	Forecast Oct-2023	Forecast Nov-2023	Forecast Dec-2023	Forecast Jan-2024	Forecast Feb-2024	Forecast Mar-2024
Dth												
Beginning Inventory	837,470	3,406,616	13,650,646	23,626,033	33,044,039	41,646,329	50,561,530	58,739,044	55,554,911	44,183,970	24,704,019	7,595,371
Injections	3,065,370	10,244,030	9,975,387	9,418,006	8,602,290	8,915,200	8,177,514	259,911	5,712	-	2,856	4,760
Withdrawals	496,225	-	-	-	-	-	-	3,444,045	11,376,652	19,479,952	17,111,504	6,917,194
EndingBalance	3,406,616	13,650,646	23,626,033	33,044,039	41,646,329	50,561,530	58,739,044	55,554,911	44,183,970	24,704,019	7,595,371	682,937
$												
Beginning Inventory	$ 1,615,813	$ 6,870,090	$ 26,571,078	$ 46,083,568	$ 64,754,437	$ 81,917,312	$ 99,253,658	$ 115,296,381	$ 109,089,458	$ 86,815,844	$ 48,448,861	$ 14,867,083
Injections	$ 6,208,024	$ 19,700,988	$ 19,512,490	$ 18,670,869	$ 17,162,876	$ 17,336,345	$ 16,042,725	$ 566,625	$ 14,367	$ -	$ 7,598	$ 11,885
Withdrawals	$ 953,747	$ -	$ -	$ -	$ -	$ -	$ -	$ 6,773,548	$ 22,287,982	$ 38,366,984	$ 33,589,376	$ 13,528,212
EndingBalance	$ 6,870,090	$ 26,571,078	$ 46,083,568	$ 64,754,437	$ 81,917,312	$ 99,253,658	$ 115,296,381	$ 109,089,458	$ 86,815,844	$ 48,448,861	$ 14,867,083	$ 1,350,756
Average Rate	$ 2.0167	$ 1.9465	$ 1.9505	$ 1.9596	$ 1.9670	$ 1.9630	$ 1.9629	$ 1.9636	$ 1.9649	$ 1.9612	$ 1.9574	$ 1.9779

Gulf Coast	Forecast Apr-2023	Forecast May-2023	Forecast Jun-2023	Forecast Jul-2023	Forecast Aug-2023	Forecast Sep-2023	Forecast Oct-2023	Forecast Nov-2023	Forecast Dec-2023	Forecast Jan-2024	Forecast Feb-2024	Forecast Mar-2024
Dth												
Beginning Inventory	2,552,927	3,078,644	3,708,450	4,052,610	4,052,610	4,052,610	4,052,610	4,052,610	4,052,610	4,052,610	4,052,610	4,052,610
Injections	525,718	629,806	344,160	-	-	-	-	-	-	-	-	-
Withdrawals	-	-	-	-	-	-	-	-	-	-	-	1,499,683
EndingBalance	3,078,644	3,708,450	4,052,610	4,052,610	4,052,610	4,052,610	4,052,610	4,052,610	4,052,610	4,052,610	4,052,610	2,552,927
$												
Beginning Inventory	$ 5,998,270	$ 7,293,549	$ 8,841,425	$ 9,702,384	$ 9,702,384	$ 9,702,384	$ 9,702,384	$ 9,702,384	$ 9,702,384	$ 9,702,384	$ 9,702,384	$ 9,702,384
Injections	$ 1,295,279	$ 1,547,875	$ 860,959	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -
Withdrawals	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ -	$ 3,590,403
EndingBalance	$ 7,293,549	$ 8,841,425	$ 9,702,384	$ 9,702,384	$ 9,702,384	$ 9,702,384	$ 9,702,384	$ 9,702,384	$ 9,702,384	$ 9,702,384	$ 9,702,384	$ 6,111,980
Average Rate	$ 2.3691	$ 2.3841	$ 2.3941	$ 2.3941	$ 2.3941	$ 2.3941	$ 2.3941	$ 2.3941	$ 2.3941	$ 2.3941	$ 2.3941	$ 2.3941

LNG	Forecast Apr-2023	Forecast May-2023	Forecast Jun-2023	Forecast Jul-2023	Forecast Aug-2023	Forecast Sep-2023	Forecast Oct-2023	Forecast Nov-2023	Forecast Dec-2023	Forecast Jan-2024	Forecast Feb-2024	Forecast Mar-2024
Dth												
Beginning Inventory	1,574,294	1,608,774	1,909,189	2,063,622	1,983,146	1,908,969	2,236,090	2,295,404	2,298,000	2,261,339	1,282,129	1,247,834
Injections	112,360	380,891	232,312	-	6,300	405,000	139,790	38,074	-	-	-	-
Withdrawals	77,880	80,476	77,880	80,476	80,476	77,880	80,476	35,478	36,661	979,210	34,295	36,661
EndingBalance	1,608,774	1,909,189	2,063,622	1,983,146	1,908,969	2,236,090	2,295,404	2,298,000	2,261,339	1,282,129	1,247,834	1,211,174
$												
Beginning Inventory	$ 2,899,161	$ 2,960,760	$ 3,481,711	$ 3,755,405	$ 3,608,822	$ 3,473,834	$ 4,064,689	$ 4,173,824	$ 4,184,763	$ 4,118,005	$ 2,326,910	$ 2,264,459
Injections	$ 202,905	$ 668,155	$ 415,590	$ -	$ 11,603	$ 732,698	$ 255,704	$ 75,487	$ -	$ -	$ -	$ -
Withdrawals	$ 141,306	$ 147,204	$ 141,896	$ 146,583	$ 146,590	$ 141,843	$ 146,570	$ 64,548	$ 66,758	$ 1,791,095	$ 62,451	$ 66,758
EndingBalance	$ 2,960,760	$ 3,481,711	$ 3,755,405	$ 3,608,822	$ 3,473,834	$ 4,064,689	$ 4,173,824	$ 4,184,763	$ 4,118,005	$ 2,326,910	$ 2,264,459	$ 2,197,701
Average Rate	$ 1.8404	$ 1.8237	$ 1.8198	$ 1.8197	$ 1.8197	$ 1.8178	$ 1.8183	$ 1.8210	$ 1.8210	$ 1.8149	$ 1.8147	$ 1.8145

Exhibit_____(EDA-1SS)
Page 4 of 4

Exhibit ___ (EDA-2SS)

KEDNY/KEDLI Forecast of Variable Gas Expense
for the TME March 31, 2021, 2022, 2023 and 2024

KEDNY / KEDLI

Forecast of Variable Gas Expense

Twelve Months Ended March 2021
(in thousands of dollars)

	Apr-2020	May-2020	Jun-2020	Jul-2020	Aug-2020	Sep-2020	Oct-2020	Nov-2020	Dec-2020	Jan-2021	Feb-2021	Mar-2021	Total TME Mar 2021
Purchased "Wellhead" Volume - MDT	21,923	22,607	18,661	16,124	16,638	18,623	23,285	19,600	22,823	22,253	21,132	22,120	245,789
Delivered Volume - MDT	21,644	22,242	18,306	15,789	16,305	18,272	22,902	19,308	22,425	21,872	20,812	21,828	241,706
HH NYMEX **(8/28/19)**	$ 2.29	$ 2.28	$ 2.32	$ 2.36	$ 2.37	$ 2.35	$ 2.37	$ 2.43	$ 2.59	$ 2.71	$ 2.68	$ 2.57	
Total "Wellhead" Cost of Purchased Gas	$ 39,594	$ 38,114	$ 31,141	$ 26,757	$ 28,093	$ 28,883	$ 38,033	$ 35,625	$ 45,070	$ 50,339	$ 47,803	$ 47,693	$ 457,145
Total Pipeline Variable Cost	$ 4,592	$ 4,846	$ 4,777	$ 4,384	$ 4,122	$ 4,206	$ 4,086	$ 3,942	$ 12,266	$ 12,533	$ 9,689	$ 7,949	$ 77,391
Total Variable Cost	$ 44,186	$ 42,960	$ 35,918	$ 31,141	$ 32,215	$ 33,089	$ 42,119	$ 39,567	$ 57,336	$ 62,872	$ 57,492	$ 55,642	$ 534,536
"Wellhead" WACOG per Dth	$ 1.81	$ 1.69	$ 1.67	$ 1.66	$ 1.69	$ 1.55	$ 1.63	$ 1.82	$ 1.98	$ 2.26	$ 2.26	$ 2.16	
Delivered Unit Cost of Total Flowing Supply Purchases	$ 2.04	$ 1.93	$ 1.96	$ 1.97	$ 1.98	$ 1.81	$ 1.84	$ 2.05	$ 2.56	$ 2.88	$ 2.76	$ 2.55	

Twelve Months Ended March 2022
(in thousands of dollars)

	Apr-2021	May-2021	Jun-2021	Jul-2021	Aug-2021	Sep-2021	Oct-2021	Nov-2021	Dec-2021	Jan-2022	Feb-2022	Mar-2022	Total TME Mar 2022
Purchased "Wellhead" Volume - MDT	22,356	22,755	18,915	17,376	16,527	18,445	23,088	19,716	23,480	22,075	21,090	22,380	248,203
Delivered Volume - MDT	22,048	22,367	18,577	17,016	16,193	18,101	22,704	19,426	23,057	21,696	20,782	22,093	244,061
HH NYMEX **(8/28/19)**	$ 2.32	$ 2.30	$ 2.33	$ 2.37	$ 2.38	$ 2.37	$ 2.40	$ 2.46	$ 2.63	$ 2.75	$ 2.73	$ 2.61	
Total "Wellhead" Cost of Purchased Gas	$ 39,365	$ 37,721	$ 30,582	$ 27,186	$ 26,909	$ 28,749	$ 37,996	$ 35,687	$ 49,171	$ 54,412	$ 51,812	$ 44,788	$ 464,377
Total Pipeline Variable Cost	$ 4,646	$ 4,908	$ 4,610	$ 4,617	$ 3,989	$ 3,828	$ 4,228	$ 4,233	$ 12,977	$ 7,150	$ 5,458	$ 8,372	$ 69,015
Total Variable Cost	$ 44,011	$ 42,628	$ 35,191	$ 31,803	$ 30,898	$ 32,577	$ 42,224	$ 39,920	$ 62,147	$ 61,562	$ 57,270	$ 53,160	$ 533,392
"Wellhead" WACOG per Dth	$ 1.77	$ 1.66	$ 1.62	$ 1.57	$ 1.63	$ 1.57	$ 1.65	$ 1.82	$ 2.10	$ 2.47	$ 2.46	$ 2.01	
Delivered Unit Cost of Total Flowing Supply Purchases	$ 2.01	$ 1.91	$ 1.90	$ 1.88	$ 1.91	$ 1.81	$ 1.87	$ 2.07	$ 2.71	$ 2.84	$ 2.76	$ 2.42	

KEDNY / KEDLI

Forecast of Variable Gas Expense

Twelve Months Ended March 2023
(in thousands of dollars)

	Apr-2022	May-2022	Jun-2022	Jul-2022	Aug-2022	Sep-2022	Oct-2022	Nov-2022	Dec-2022	Jan-2023	Feb-2023	Mar-2023	Total TME Mar 2023
Purchased "Wellhead" Volume - MDT	22,682	23,047	19,152	17,667	16,740	18,953	23,485	20,036	23,458	22,776	21,516	23,373	252,884
Delivered Volume - MDT	22,392	22,655	18,811	17,319	16,422	18,601	23,101	19,739	23,048	22,399	21,198	23,075	248,759
HH NYMEX **(8/28/19)**	$ 2.37	$ 2.36	$ 2.39	$ 2.42	$ 2.43	$ 2.43	$ 2.45	$ 2.51	$ 2.69	$ 2.81	$ 2.78	$ 2.67	
Total "Wellhead" Cost of Purchased Gas	$ 39,611	$ 37,534	$ 30,666	$ 27,855	$ 27,001	$ 29,809	$ 38,769	$ 36,575	$ 50,082	$ 54,604	$ 49,820	$ 46,120	$ 468,445
Total Pipeline Variable Cost	$ 4,725	$ 4,916	$ 4,797	$ 4,445	$ 3,955	$ 3,890	$ 4,309	$ 4,318	$ 13,326	$ 11,788	$ 10,500	$ 8,694	$ 79,665
Total Variable Cost	$ 44,337	$ 42,451	$ 35,463	$ 32,299	$ 30,956	$ 33,700	$ 43,078	$ 40,893	$ 63,407	$ 66,392	$ 60,320	$ 54,814	$ 548,110
"Wellhead" WACOG per Dth	$ 1.76	$ 1.64	$ 1.61	$ 1.59	$ 1.62	$ 1.58	$ 1.66	$ 1.84	$ 2.15	$ 2.41	$ 2.33	$ 1.99	
Delivered Unit Cost of Total Flowing Supply Purchases	$ 1.99	$ 1.88	$ 1.90	$ 1.88	$ 1.90	$ 1.82	$ 1.88	$ 2.09	$ 2.77	$ 2.98	$ 2.86	$ 2.40	

Twelve Months Ended March 2024
(in thousands of dollars)

	Apr-2023	May-2023	Jun-2023	Jul-2023	Aug-2023	Sep-2023	Oct-2023	Nov-2023	Dec-2023	Jan-2024	Feb-2024	Mar-2024	Total TME Mar 2024
Purchased "Wellhead" Volume - MDT	23,534	23,812	19,765	18,011	17,088	19,347	24,037	20,218	23,990	22,413	22,741	24,359	259,314
Delivered Volume - MDT	23,214	23,408	19,405	17,662	16,764	18,990	23,643	19,918	23,573	22,039	22,397	24,056	255,070
HH NYMEX **(8/28/19)**	$ 2.45	$ 2.44	$ 2.48	$ 2.53	$ 2.54	$ 2.54	$ 2.56	$ 2.63	$ 2.80	$ 2.92	$ 2.89	$ 2.78	
Total "Wellhead" Cost of Purchased Gas	$ 41,031	$ 39,294	$ 32,102	$ 29,027	$ 28,238	$ 31,216	$ 40,869	$ 38,666	$ 53,614	$ 54,375	$ 55,715	$ 50,788	$ 494,934
Total Pipeline Variable Cost	$ 5,262	$ 5,223	$ 5,052	$ 4,619	$ 4,090	$ 4,174	$ 4,527	$ 4,541	$ 14,318	$ 10,492	$ 11,565	$ 9,404	$ 83,266
Total Variable Cost	$ 46,293	$ 44,517	$ 37,153	$ 33,646	$ 32,327	$ 35,390	$ 45,396	$ 43,207	$ 67,931	$ 64,867	$ 67,280	$ 60,192	$ 578,200
"Wellhead" WACOG per Dth	$ 1.76	$ 1.66	$ 1.64	$ 1.63	$ 1.67	$ 1.63	$ 1.71	$ 1.93	$ 2.26	$ 2.45	$ 2.47	$ 2.11	
Delivered Unit Cost of Total Flowing Supply Purchases	$ 2.01	$ 1.92	$ 1.93	$ 1.92	$ 1.94	$ 1.88	$ 1.94	$ 2.19	$ 2.92	$ 2.97	$ 3.03	$ 2.53	

Second Supplemental Direct Testimony of Elizabeth D. Arangio

Exhibit ___ (EDA-3SS)

KEDNY/KEDLI Forecast of Purchased Gas Expense
for the TME March 31, 2021, 2022, 2023 and 2024

KEDNY / KEDLI

Forecast of Purchased Gas Expense

Twelve Months Ended March 2021
(in thousands of dollars)

	Apr-2020	May-2020	Jun-2020	Jul-2020	Aug-2020	Sep-2020	Oct-2020	Nov-2020	Dec-2020	Jan-2021	Feb-2021	Mar-2021	Total TME Mar 2021
Delivered Volume - MDT	21,644	22,242	18,306	15,789	16,305	18,272	22,902	19,308	22,425	21,872	20,812	21,828	241,706
Variable Cost	$ 44,186	$ 42,960	$ 35,918	$ 31,141	$ 32,215	$ 33,089	$ 42,119	$ 39,567	$ 57,336	$ 62,872	$ 57,492	$ 55,642	$ 534,536
Fixed Costs	$ 42,089	$ 42,089	$ 42,089	$ 42,089	$ 42,089	$ 42,089	$ 42,089	$ 42,089	$ 50,882	$ 50,882	$ 50,882	$ 42,089	$ 531,443
Total Invoice Cost	$ 86,275	$ 85,049	$ 78,007	$ 73,229	$ 74,303	$ 75,177	$ 84,208	$ 81,655	$ 108,218	$ 113,754	$ 108,374	$ 97,730	$ 1,065,980
Minus Injections to Storage	$ (6,673)	$ (22,489)	$ (21,103)	$ (17,227)	$ (18,269)	$ (17,882)	$ (16,368)	$ (1,495)	$ (13)	$ -	$ -	$ (11)	$ (121,531)
Plus Withdrawls from Storage	$ 727	$ 159	$ 153	$ 157	$ 157	$ 150	$ 153	$ 6,979	$ 22,641	$ 38,396	$ 31,329	$ 19,606	$ 120,608
Total Purchased Gas Expense	$ 80,329	$ 62,719	$ 57,057	$ 56,160	$ 56,192	$ 57,446	$ 67,994	$ 87,139	$ 130,846	$ 152,150	$ 139,703	$ 117,325	$ 1,065,057
WACOG per Dth													
Delivered Unit Cost of Total Flowing Supply Purchases	$ 2.04	$ 1.93	$ 1.96	$ 1.97	$ 1.98	$ 1.81	$ 1.84	$ 2.05	$ 2.56	$ 2.87	$ 2.76	$ 2.55	
Underground Storage "In Ground" WACOG	$ 2.23	$ 2.09	$ 2.06	$ 2.05	$ 2.05	$ 2.02	$ 2.00	$ 2.00	$ 2.01	$ 2.02	$ 2.09	$ 2.25	
LNG WACOG	$ 1.99	$ 1.96	$ 1.95	$ 1.95	$ 1.95	$ 1.91	$ 1.89	$ 1.89	$ 1.89	$ 1.89	$ 1.89	$ 1.89	

Twelve Months Ended March 2022
(in thousands of dollars)

	Apr-2021	May-2021	Jun-2021	Jul-2021	Aug-2021	Sep-2021	Oct-2021	Nov-2021	Dec-2021	Jan-2022	Feb-2022	Mar-2022	Total TME Mar 2022
Delivered Volume - MDT	22,048	22,367	18,577	17,016	16,193	18,101	22,704	19,426	23,057	21,696	20,782	22,093	244,061
Variable Cost	$ 44,011	$ 42,628	$ 35,191	$ 31,803	$ 30,898	$ 32,577	$ 42,224	$ 39,920	$ 62,147	$ 61,562	$ 57,270	$ 53,160	$ 533,392
Fixed Costs	$ 42,089	$ 42,089	$ 42,089	$ 42,089	$ 42,089	$ 42,089	$ 42,089	$ 42,089	$ 76,034	$ 76,034	$ 76,034	$ 42,089	$ 606,900
Total Invoice Cost	$ 86,100	$ 84,717	$ 77,280	$ 73,891	$ 72,987	$ 74,666	$ 84,312	$ 82,009	$ 138,181	$ 137,596	$ 133,304	$ 95,248	$ 1,140,292
Minus Injections to Storage	$ (6,770)	$ (21,801)	$ (20,342)	$ (18,211)	$ (17,063)	$ (17,225)	$ (15,636)	$ (732)	$ (7)	$ -	$ -	$ (11)	$ (117,798)
Plus Withdrawls from Storage	$ 762	$ 152	$ 147	$ 152	$ 152	$ 146	$ 150	$ 6,488	$ 21,633	$ 38,618	$ 31,443	$ 19,237	$ 119,081
Total Purchased Gas Expense	$ 80,091	$ 63,069	$ 57,084	$ 55,832	$ 56,076	$ 57,588	$ 68,826	$ 87,765	$ 159,808	$ 176,214	$ 164,748	$ 114,474	$ 1,141,574
WACOG per Dth													
Delivered Unit Cost of Total Flowing Supply Purchases	$ 2.00	$ 1.91	$ 1.89	$ 1.87	$ 1.91	$ 1.80	$ 1.86	$ 2.05	$ 2.70	$ 2.84	$ 2.76	$ 2.41	
Underground Storage "In Ground" WACOG	$ 2.18	$ 2.06	$ 2.01	$ 1.99	$ 1.99	$ 1.97	$ 1.96	$ 1.96	$ 1.97	$ 1.99	$ 2.06	$ 2.24	
LNG WACOG	$ 1.89	$ 1.88	$ 1.88	$ 1.88	$ 1.88	$ 1.86	$ 1.86	$ 1.86	$ 1.86	$ 1.87	$ 1.87	$ 1.87	

KEDNY / KEDLI

Forecast of Purchased Gas Expense

Twelve Months Ended March 2023
(in thousands of dollars)

	Apr-2022	May-2022	Jun-2022	Jul-2022	Aug-2022	Sep-2022	Oct-2022	Nov-2022	Dec-2022	Jan-2023	Feb-2023	Mar-2023	Total TME Mar 2023
Delivered Volume - MDT	22,392	22,655	18,811	17,319	16,422	18,601	23,101	19,739	23,048	22,399	21,198	23,075	248,759
Variable Cost	$ 44,337	$ 42,451	$ 35,463	$ 32,299	$ 30,956	$ 33,700	$ 43,078	$ 40,893	$ 63,407	$ 66,392	$ 60,320	$ 54,814	$ 548,110
Fixed Costs	$ 42,089	$ 42,089	$ 42,089	$ 42,089	$ 42,089	$ 42,089	$ 42,089	$ 42,089	$ 72,669	$ 72,669	$ 72,669	$ 42,089	$ 596,804
Total Invoice Cost	$ 86,425	$ 84,539	$ 77,551	$ 74,388	$ 73,045	$ 75,788	$ 85,166	$ 82,982	$ 136,076	$ 139,061	$ 132,989	$ 96,903	$ 1,144,914
Minus Injections to Storage	$ (6,745)	$ (21,176)	$ (20,118)	$ (18,244)	$ (16,744)	$ (17,614)	$ (15,738)	$ (703)	$ (13)	$ -	$ -	$ (9)	$ (117,106)
Plus Withdrawls from Storage	$ 738	$ 127	$ 122	$ 126	$ 126	$ 121	$ 125	$ 6,421	$ 22,126	$ 37,855	$ 31,271	$ 17,703	$ 116,859
Total Purchased Gas Expense	$ 80,418	$ 63,490	$ 57,555	$ 56,269	$ 56,426	$ 58,295	$ 69,553	$ 88,700	$ 158,188	$ 176,916	$ 164,260	$ 114,597	$ 1,144,668
WACOG per Dth													
Delivered Unit Cost of Total Flowing Supply Purchases	$ 1.98	$ 1.87	$ 1.89	$ 1.86	$ 1.89	$ 1.81	$ 1.86	$ 2.07	$ 2.75	$ 2.96	$ 2.85	$ 2.38	
Underground Storage "In Ground" WACOG	$ 2.18	$ 2.02	$ 1.99	$ 1.98	$ 1.97	$ 1.96	$ 1.95	$ 1.95	$ 1.96	$ 1.98	$ 2.06	$ 2.25	
LNG WACOG	$ 1.87	$ 1.86	$ 1.86	$ 1.86	$ 1.86	$ 1.84	$ 1.84	$ 1.84	$ 1.84	$ 1.84	$ 1.84	$ 1.84	

Twelve Months Ended March 2024
(in thousands of dollars)

	Apr-2023	May-2023	Jun-2023	Jul-2023	Aug-2023	Sep-2023	Oct-2023	Nov-2023	Dec-2023	Jan-2024	Feb-2024	Mar-2024	Total TME Mar 2024
Delivered Volume - MDT	22,392	22,655	18,811	17,319	16,422	18,601	23,101	19,739	23,048	22,399	21,198	23,075	248,759
Variable Cost	$ 44,337	$ 42,451	$ 35,463	$ 32,299	$ 30,956	$ 33,700	$ 43,078	$ 40,893	$ 63,407	$ 66,392	$ 60,320	$ 54,814	$ 548,110
Fixed Costs	$ 42,135	$ 42,135	$ 42,135	$ 42,135	$ 42,135	$ 42,135	$ 42,135	$ 42,135	$ 72,938	$ 72,938	$ 72,938	$ 42,135	$ 598,029
Total Invoice Cost	$ 86,471	$ 84,586	$ 77,598	$ 74,434	$ 73,091	$ 75,834	$ 85,213	$ 83,028	$ 136,346	$ 139,331	$ 133,259	$ 96,949	$ 1,146,139
Minus Injections to Storage	$ (7,706)	$ (21,917)	$ (20,789)	$ (18,671)	$ (17,174)	$ (18,069)	$ (16,298)	$ (642)	$ (14)	$ -	$ (8)	$ (12)	$ (121,301)
Plus Withdrawls from Storage	$ 1,095	$ 147	$ 142	$ 147	$ 147	$ 142	$ 147	$ 6,838	$ 22,355	$ 40,158	$ 33,652	$ 17,185	$ 122,154
Total Purchased Gas Expense	$ 79,860	$ 62,816	$ 56,950	$ 55,910	$ 56,063	$ 57,907	$ 69,061	$ 89,224	$ 158,686	$ 179,489	$ 166,903	$ 114,122	$ 1,146,992
WACOG per Dth													
Delivered Unit Cost of Total Flowing Supply Purchases	$ 1.98	$ 1.87	$ 1.89	$ 1.86	$ 1.89	$ 1.81	$ 1.86	$ 2.07	$ 2.75	$ 2.96	$ 2.85	$ 2.38	
Underground Storage "In Ground" WACOG	$ 2.18	$ 2.04	$ 2.02	$ 2.01	$ 2.00	$ 2.00	$ 1.99	$ 1.99	$ 2.00	$ 2.02	$ 2.11	$ 2.31	
LNG WACOG	$ 1.84	$ 1.82	$ 1.82	$ 1.82	$ 1.82	$ 1.82	$ 1.82	$ 1.82	$ 1.82	$ 1.81	$ 1.81	$ 1.81	

Second Supplemental Direct Testimony of Elizabeth D. Arangio

Exhibit ___ (EDA-4SS)

KEDNY and KEDLI Estimated Marginal Commodity Cost of Gas

KEDNY
Estimated Marginal Commodity Cost of Gas
For Period: November 1, 2020 through March 31, 2021
($ / dt)

Nov	Dec	Jan	Feb	Mar	Winter (Nov-Mar) Average
$ 2.39	$ 5.23	$ 8.46	$ 5.76	$ 4.70	**$ 5.31**

KEDLI
Estimated Marginal Commodity Cost of Gas
For Period: November 1, 2020 through March 31, 2021
($ / dt)

Nov	Dec	Jan	Feb	Mar	Winter (Nov-Mar) Average
$ 2.39	$ 5.23	$ 8.46	$ 5.76	$ 4.70	**$ 5.31**

Second Supplemental Direct Testimony of Elizabeth D. Arangio

Exhibit __ (EDA-5SS)

KEDNY and KEDLI Estimated Annualized Marginal Capacity Cost of Gas

KEDNY
Estimated Annualized Marginal Capacity Cost of Gas
For Period: November 1, 2020 through March 31, 2021
Units ($ per dt)

Marginal Supplies	Peak Day Quantity dt/day		Annual Capacity Costs Cost $	Quantity dt	Unitized $/dt	Peak Day Capacity Costs $/dt
CNG Peaking Services	8,800	$	[redacted]	132,000	[redacted]	[redacted]
15 Day City Gate Peaking Supplies	221,000	$	[redacted]	3,315,000	[redacted]	[redacted]
30 Day City Gate Peaking Supplies	79,000	$	[redacted]	2,370,000	[redacted]	[redacted]
Grand Total	308,800		[redacted]	5,817,000	[redacted]	[redacted]

Annualized Marginal Capacity Cost of Gas

Peak Day Capacity Cost	[redacted]	**per dt**
Ratio: Peak Day Requirements to Annual Normalized Firm Sales	**1 to 80**	**dt**
Annual Marginal Capacity Cost	[redacted]	**per dt**

KEDLI
Estimated Annualized Marginal Capacity Cost of Gas
For Period: November 1, 2020 through March 31, 2021
Units ($ per dt)

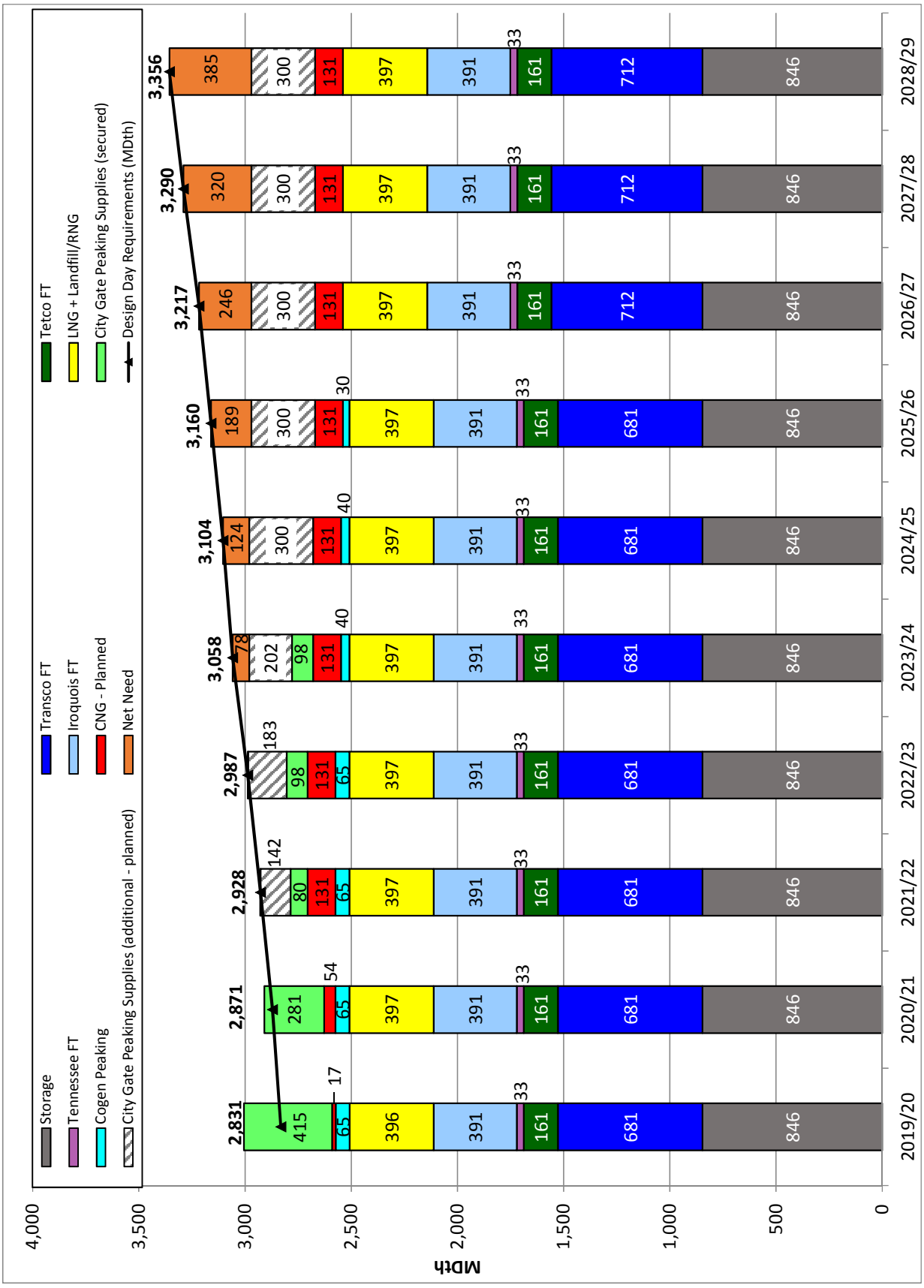
Marginal Supplies	Peak Day Quantity dt/day		Annual Capacity Costs Cost $	Quantity dt	Unitized $/dt	Peak Day Capacity Costs $/dt
CNG Peaking Services	8,800	$	[redacted]	132,000	[redacted]	[redacted]
15 Day City Gate Peaking Supplies	221,000	$	[redacted]	3,315,000	[redacted]	[redacted]
30 Day City Gate Peaking Supplies	79,000	$	[redacted]	2,370,000	[redacted]	[redacted]
Grand Total	308,800		[redacted]	5,817,000	[redacted]	[redacted]

Annualized Marginal Capacity Cost of Gas

Peak Day Capacity Cost	[redacted]	**per dt**
Ratio: Peak Day Requirements to Annual Normalized Firm Sales	**1 to 80**	**dt**
Annual Marginal Capacity Cost	[redacted]	**per dt**

Exhibit __ (EDA-6SS)

The Companies' updated peak day supply/demand


Storage
Tennessee FT
Cogen Peaking
City Gate Peaking Supplies (additional - planned)
Transco FT
Iroquois FT
CNG - Planned
Net Need
Tetco FT
LNG + Landfill/RNG
City Gate Peaking Supplies (secured)
Design Day Requirements (MDth)
MDth
4,000
3,500
3,000
2,500
2,000
1,500
1,000
500
0
2019/20
2020/21
2021/22
2022/23
2023/24
2024/25
2025/26
2026/27
2027/28
2028/29
2,831
2,871
2,928
2,987
3,058
3,104
3,160
3,217
3,290
3,356